INCH-POUND

MIL-T-5790F(USAF)

30 JUNE 1994

SUPERSEDING
MIL-T-5790E(USAF)
04 JULY 1988

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, OIL, SYNCHRO, 0-200 PSI 320-DEGREE MOVEMENT, 26V, SINGLE PHASE, 400 CYCLES

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating 320-degree movement, synchro style, differential, oil pressure transmitter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Oklahoma City Air Logistics center/TICLA, Tinker AFB OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-5350	Transmitters, Synchro Operated, Aircraft, General Specification For
MIL-I-7087	Indicator, Pressure, Oil, 0-200 PSI

STANDARDS

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-2073-1	DOD Material Procedure For Development And Application Of Packaging Requirements
MS28005	Transmitters, Pressure, Synchro Aircraft
MS28010	Indicators, Pressure-Synchro, Single, 2-Inch Size

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing and Materials (ASTM)
ASTM D3951 Packaging Commercial
ASTM D4727 Standard Specification For Corrugated And Solid Fiberboard Sheet Stock (Container Grade) And Cut Shapes.

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The transmitter furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2).

3.2 General. The requirements specified in MIL-T-5350 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.3. Design. The oil pressure transmitter shall be design for use with an MS28010-3 pressure indicator conforming to MIL-I-7087.

3.3.1 Hermetically sealed case. The case shall provide a Hermetically sealed enclosure for the entire internal mechanism of the transmitter. The design of the case shall permit removal and replacement of the internal mechanism and resealing of the case. This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will no be dependent upon materials which will be affected by the action of any atmosphere to which the transmitter may be subjected.

3.4 Construction.

3.4.1 Dimensions. The outline dimensions of the transmitter shall conform to Standard MS28005.

3.4.2 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the pointer of any indicator to travel beyond full scale to a point approximately midway between the last and the zero graduation.

3.4.3 Bosses. The bosses, as shown on MS28005, shall be provided for external pressure and vent connections to the pressure sensitive element and the opposing pressure element, respectively. The bosses shall be identified with white lettering "P" and "V" permanently inscribed 0.125 inch in height.

3.4.4 Restrictors. No restrictors or capillary line connecting the pressure fitting to the pressure element shall be used which will introduce an objectionable lag in the indication response.

3.4 Weight. The weight of the transmitter shall not exceed 1.4 pounds.

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The pressure range of the transmitter shall be included as Special Characteristics as specified in MIL-STD-130.

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3.7 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.8 Cadmium. Cadmium shall not be used on or in the transmitter without written approval of the preparing activity. Cadmium shall not be used on the exterior of the transmitter. Cadmium shall not come into contact with the fluid being measured.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.5).
b. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in this specification.

4.4 Tolerances. When the transmitter is subjected to the tests specified in MIL-T-5350, the tolerances shall not exceed those specified in tables I and II.

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4.4.1 Scale error. When testing or scale error in accordance with MIL-T-5350, the transmitter shall be tested first with

increasing readings and then with decreasing readings. With the readings increasing, the indication shall be brought up to, but no exceeding, the indication specified to give the desired readings. With readings decreasing, the indications shall be brought down to, but not fall below, the indication specified to give the desired reading.

TABLE I
Tolerances (psi)

Test (psi)	Room temperature	Vibration, overpressure	Low temperature, high temperature, seasoning endurance, and altitude tests
0	3.0	4.0	5.0
50	3.0	4.0	5.0
100	4.0	5.0	6.0
150	4.0	5.0	6.0
200	4.0	5.0	6.0

TABLE II
Oil pressure transmitter

Test	Tolerance-(psi)
Friction:	
Room temperature	3.0
High temperature	5.0
Cold temperature	5.0
Position error at 100 psi	1.0
Vibration	
Pointer oscillation	3.0

4.5 Qualification inspection. Three qualification inspection samples representative of production item shall be inspected in accordance to paragraph 4.5 of MIL-T-5350 and as specified herein.

4.6 Quality conformance inspection. Quality conformance inspection shall be accomplished as called out in MIL-T-5350.

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4.7 Special tests.

4.7.1 Overpressure. The transmitter shall be subjected to a pressure of 400 pounds per square inch (psi) for a period of 10 minutes. The transmitter shall be tested not less than 1 hour following the Overpressure test for scale error as specified under individual tests on increasing pressure only. The scale error shall not exceed the tolerance specified in table I.

4.7.2 Damping. A pressure sufficient to produce full scale pointer deflection shall be applied to the transmitter. A quick release valve shall be used to exhaust the pressure. The time required for the pointer to traverse the interval from 200 psi to 20 psi shall be within the period of 9 ± 3 seconds. The test shall be conducted with lubricating oil with a viscosity of approximately 2,500 centistokes, such as grade 1005 oil at -65°F, or equivalent.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaning in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1A8 of MIL-P-116. Each transmitter shall be packed in Fiberboard shipping containers conforming to PPP-B-636, Box Shipping, Fiberboard or ASTM D 4727, Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes. Sufficient cushioning material between the bag and unit container shall be provided. The cushioning material shall be of the proper type, density, and thickness to ensure shock transmission does not exceed peak values in G's established for the compass when completed packs are subject to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

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5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in Fiberboard shipping containers conforming to PPP-B-636 or ASTM D4727, unless otherwise specified by the contracting activity. All other requirements, as specified by the contracting activity, shall apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oil pressure transmitter covered by this specification is intended to transmit the oil pressure of the lubricating system of aircraft engines to an applicable indicator.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Applicable level of packaging and packing (see 5.1).
(c) With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the Qualified Products List (QPL) 5790 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal

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Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders from the Products List is OC-ALC/TICLA Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

(d) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Subject term (key word) listing.
Oil
Pressure
Synchro
Transmitter
10G
26 VAC
0-200 PSI

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
AIR FORCE -99

Preparing activity:
AIR FORCE -71

Project number;
6620-F545

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-5790F(USAF)	94/06/30

3. DOCUMENT TITLE
Transmitter, Pressure, Oil, Synchro, 0-200 PSI 320°

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME OC-ALC/TICLA	b. TELEPHONE (Include Area Code) (1) Commercial (405) 736-5960 (2) AUTOVON
c. ADDRESS (Include Zip Code) 3001 Staff Dr. Tinker AFB, OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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